

Q1. Write a Python program to create two DataFrames, student\_details and academic\_records, containing suitable student-related attributes. Perform data integration by combining the two DataFrames using a common attribute such as Student\_ID. Display the individual DataFrames and the integrated DataFrame.

[10 Marks]

Q2. Create a dataset containing information about different cities with the attributes: City, Population, GDP, and Literacy Rate. Perform the following visualization tasks:

- a. Generate a Treemap showing city-wise population distribution.
- b. Create a 3D Scatter Plot using Population, GDP, and Literacy Rate.

[20 Marks]

**OR**

Q2. Develop a Random Forest Classifier using the given Student Grading dataset to classify students into different performance categories. Perform the necessary data preprocessing, model training, and testing, and evaluate the classifier using appropriate classification metrics such as Confusion matrix, Accuracy, Precision, Recall, and F1-Score.

[20 Marks]

Q3. Viva

[5 Marks]